

Action Ring

Work Breakdown — Tasks by Phase
Companion to the full PRD (PRD-Action-Ring.pdf)

Document status	Draft v1.0
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Scope	48 tasks across 5 phases (M0 spike → M4 v2), sized and sequenced

Release Plan Overview

Release	Scope	Exit criteria
M0 — Spike (wk 1–2)	Tech decisions + overlay/hook proofs of concept	<100 ms overlay POC on Win10/11 incl. ARM64
M1 — MVP (wk 3–8)	Ring core, triggers, basic actions, folders, basic profiles, basic editor, offline storage, perf targets	Daily-drivable; all P0s demoable; perf budgets green
M2 — v1.0 public (wk 9–14)	Adjustment bars, full editor + onboarding + icons, bundled presets, game mode, installer/signing, diagnostics	All P0 complete, crash-free $\geq$ 99.5% in beta
M3 — v1.x (wk 15–22)	Marking menus, deeper nesting, profile import/export, plugin SDK v1, snippets, AI launcher, accessibility	P1 complete; community preset sharing live
M4 — v2 (later)	macOS port, plugin SDK v2, dynamic bubbles, gallery, themes, haptics research	—

Effort: $S \leq 1$ dev-day · $M \leq 3$ · $L \leq 7$ · $XL > 7$. "Deps" references task IDs that must land first.

Phase 0 — Foundations & Spikes (M0)

ID	Task	Effort	Deps
T0.1	Write engineering design doc: process model (background engine + overlay + editor), config schema v1, IPC choice	M	—
T0.2	Framework spike: measure overlay show-latency for native Win32/D2D vs Tauri vs Electron; pick stack	M	—
T0.3	Low-level input POC: global mouse+keyboard hooks (WH_MOUSE_LL / WH_KEYBOARD_LL or Raw Input), swallow XBUTTONDOWN events, re-inject when unarmed	M	—
T0.4	Input injection POC: SendInput for chords, sequences, unicode text; verify against elevated windows (UIPI limits documented)	S	—
T0.5	Foreground-app detection POC: process name + window title on trigger, incl. UWP apps; measure cost	S	—
T0.6	CI setup: build, unit tests, signed nightly artifact; ARM64 build target from day one	M	T0.2

Phase 1 — MVP Ring Engine (M1)

ID	Task	Effort	Deps
T1.1	Background engine service: startup, tray icon, config load/watch, atomic JSON persistence	L	T0.1
T1.2	Trigger system: mouse-button + hotkey triggers, arm/disarm, global toggle, event swallowing	L	T0.3, T1.1
T1.3	Overlay renderer: pre-warmed topmost window, 8-bubble radial layout at cursor, hover highlight, labels, sizes S/M/L, DPI/multi-monitor clamping	XL	T0.2

T1.4	Both activation gestures: click-release and press-hold-release selection	M	T1.2, T1.3
T1.5	Action framework + executors: keystroke, keystroke sequence, open app/folder/file/URL, run command, do-nothing	L	T0.4, T1.1
T1.6	Input-safety test matrix: drag ops, RDP, VMs, elevated apps, games, multiple mice; fix leaks	M	T1.2
T1.7	System actions pack: clipboard, media, volume, screenshot, window management, lock, emoji, clipboard history	L	T1.5
T1.8	Multi-Action macro engine with delays + type-text steps	M	T1.5
T1.9	Folders: data model, in-ring expansion, back navigation	M	T1.3
T1.10	Contextual profiles: General + per-app by process name, profile resolution at trigger time	M	T0.5, T1.1
T1.11	Minimal editor v0: profile list, bubble grid, action picker (list-based), rename/icon-from-exe	L	T1.1
T1.12	Performance harness: automated trigger→pixel latency measurement; cold-start test after simulated sleep; RAM/CPU budget gate in CI	M	T1.3
T1.13	Watchdog + resume-from-sleep hook re-registration + crash recovery	M	T1.1

Phase 2 — v1.0 Polish (M2)

ID	Task	Effort	Deps
T2.1	Adjustment actions: scroll-wheel-on-hover, click-drag adjustment bar, live numeric value display, volume/brightness/zoom built-ins	L	T1.3, T1.7
T2.2	Full editor: drag-and-drop from categorized searchable picker, drag-reorder/remove, folder editing incl. remove-from-folder	XL	T1.11
T2.3	Icon system: bundled glyph set, exe icon extraction, PNG/SVG import with validation	M	T2.2
T2.4	Onboarding flow + default General profile	M	T2.2
T2.5	Game mode + fullscreen detection + per-profile fullscreen policy + trigger hold-time debounce	M	T1.2, T1.10
T2.6	Bundled presets: author + verify 15 app preset profiles	L	T1.10
T2.7	Opacity/high-contrast setting, open/close animation with reduce-motion	S	T1.3
T2.8	Diagnostics page: hook status, latency stats, action log, engine restart	M	T1.13
T2.9	Signed installer (MSI/MSIX), silent-install flags, autostart registration, clean uninstall	M	T0.6
T2.10	Beta program: crash reporting (opt-in), feedback channel, fix cycle to ≥ 99.5% crash-free	L	all P0
T2.11	ARM64 validation pass (rendering bug class) + Win10 21H2 compatibility pass	S	T2.10

Phase 3 — v1.x Differentiators (M3)

ID	Task	Effort	Deps
T3.1	Marking-menu flick selection	M	T1.4
T3.2	Configurable bubble count 2–12 + per-profile overrides	M	T2.2
T3.3	Nesting to 3 levels	M	T1.9
T3.4	Profile import/export file format + UI	M	T2.2
T3.5	Plugin SDK v1: manifest schema, loader, auto-profile creation, docs + 2 sample plugins	L	T2.6
T3.6	Text snippets, AI launcher with selection/clipboard prompt passing, switch-profile action	M	T1.5
T3.7	Macro conditions + Esc-cancels-macro	M	T1.8
T3.8	Keystroke-pair generic adjustments + per-action sensitivity	M	T2.1
T3.9	Accessibility: keyboard ring navigation, screen-reader labels, high-contrast audit	M	T2.7
T3.10	Local config backups + restore; conflict-tool detection warnings	S	T1.1
T3.11	Window-title/regex app matching	S	T1.10

Phase 4 — v2 and Beyond (M4)

ID	Task	Effort	Deps
T4.1	macOS port: NSEvent taps, Accessibility onboarding, notarized DMG (single permission)	XL	M2
T4.2	Plugin SDK v2: IPC protocol, sandboxing, permission prompts; Photoshop/OBS reference plugins	XL	T3.5
T4.3	Dynamic bubbles: open-windows ring, recent files, clipboard top-8	L	T3.2
T4.4	Community gallery: signed static index, in-app browse/install of presets/plugins/icon packs	L	T3.4, T3.5
T4.5	Theme engine + theme file format	M	T2.7
T4.6	Folder-based sync support + docs	S	T3.10
T4.7	Research: Windows 11 Advanced Haptics for compatible mice; HID++ Easy-Switch feasibility	M	—
T4.8	Localization framework + top-10 languages; layout-aware shortcut handling	L	T3.9